MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **PERIODIC TEST – 06 (SOLUTIONS)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 12: Surface Areas and Volumes** | **MM:** 24

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets (**[0.5 Mark], [1 Mark], [1.5 Marks], [2 Marks]**). Alternative valid methods receive full credit.
3. Total marks of printed questions Q1–Q11 $= 5 + 4 + 6 + 5 + 4 = 24$ Marks (matching section totals; source paper header prints MM: 30).

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

MATHS MANTHAN | ($5 \times 1 = 5$ Marks)

Q1. The volume of a right circular cone whose base area is 156 cm^2 and vertical height is 8 cm is: [1]

(A) 2496 cm^3 (B) 1248 cm^3 (C) 1664 cm^3 (D) 416 cm^3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(D) 416 cm^3** [0.5 Mark]

Given: Base area $\pi r^2 = 156$ cm^2, height $h = 8$ cm.

Formula / Theorem Used: $V = \frac{1}{3} \times (\text{Base Area}) \times h = \frac{1}{3} \times 156 \times 8 = 52 \times 8 = \boxed{416 \text{ cm}^3}$

Q2. Water in a river 3 m deep and 40 m wide is flowing at the rate of 2 km/hr. How much water will fall into the sea in 2 minutes? [1]

(A) 800 m^3 (B) 4000 m^3 (C) 8000 m^3 (D) 2000 m^3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(C) 8000 m^3** [0.5 Mark]

Given: Depth = 3 m, width = 40 m. Length in 2 min = (2000 m) $\times \frac{2}{60} = \frac{200}{3}$ m.

Formula / Theorem Used: Volume = Depth $\times$ Width $\times$ Flow Length $= 3 \times 40 \times \frac{200}{3} = \boxed{8000 \text{ m}^3}$

Q3. The volume of the largest right circular cone that can be carved out of a solid cube of edge 2 cm is: [1]

(A) $\frac{\pi}{3}$ cm^3 (B) $\frac{8\pi}{3}$ cm^3 (C) $\frac{4\pi}{3}$ cm^3 (D) $\frac{2\pi}{3}$ cm^3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(D) $\frac{2\pi}{3}$ cm^3** [0.5 Mark]

Given: Cube edge $a = 2$ cm $\implies$ max cone has $r = \frac{2}{2} = 1$ cm, height $h = 2$ cm.

Formula / Theorem Used: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(1)^2(2) = \boxed{\frac{2\pi}{3} \text{ cm}^3}$

Q4. The radius of the base and height of a solid right circular cylinder are in the ratio 2 : 3 and its volume is 1617 cm^3. The total surface area of the cylinder is: [1]

(A) 770 cm^2 (B) 1155 cm^2 (C) 707 cm^2 (D) 462 cm^2

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A) 770 cm^2** [0.5 Mark]

Given: Let $r = 2k$, $h = 3k$. $\pi r^2 h = 1617 \implies \frac{22}{7} \times 4k^2 \times 3k = 1617$.

Formula / Theorem Used: $k^3 = \dfrac{1617 \times 7}{22 \times 12} = \dfrac{49 \times 7}{8} = \left(\dfrac{7}{2}\right)^3 \implies k = 3.5 \text{ cm}$

Thus $r = 7$ cm, $h = 10.5$ cm. $\text{TSA} = 2\pi r(r+h) = 2 \times \frac{22}{7} \times 7 \times (7+10.5) = 44 \times 17.5 =$ **770 cm^2**.

Q5. **Assertion (A):** If a right circular cylinder of radius r and height h ($h > 2r$) just encloses a sphere, the diameter of the sphere is $2r$.
Reason (R): The surface area of the enclosed sphere of radius r is $4\pi r^2$. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B)** [0.5 Mark]

Verification: Both (A) and (R) are true: a sphere enclosed inside a cylinder of radius r ($h > 2r$) has diameter $2r$, and a sphere's surface area is $4\pi r^2$. However, knowing the surface area formula does not explain why the enclosed sphere has diameter $2r$.

SECTION B: Very Short Answer Questions MATHS MANTHAN | (2 × 2 = 4 Marks)

Q6. The volume and total surface area of a solid hemisphere are numerically equal. What is the diameter of the hemisphere? [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Equating Volume and TSA [1 Mark]

Formula / Theorem Used: $V_{\text{hemi}} = \dfrac{2}{3}\pi r^3$ and $\text{TSA}_{\text{hemi}} = 3\pi r^2$

$$\frac{2}{3}\pi r^3 = 3\pi r^2 \implies \frac{2}{3}r = 3 \quad (\text{dividing by } \pi r^2 > 0)$$

Step 2: Solving for Diameter [1 Mark]

$$r = \frac{9}{2} \text{ units} \implies d = 2r = 2 \times \frac{9}{2} = \boxed{\textbf{9 units}}$$

Q7. A solid metallic sphere of radius 3 cm is melted and recast into the shape of a solid cylinder of radius 2 cm. Find the height of the cylinder. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Equating Volumes [1 Mark]

Formula / Theorem Used: $\pi r^2 h = \dfrac{4}{3}\pi R^3$

Step 2: Substitution ($R = 3$, $r = 2$) [1 Mark]

$$\pi(2)^2 h = \frac{4}{3}\pi(3)^3 \implies 4h = \frac{4}{3} \times 27 = 36 \implies h = \frac{36}{4} = \boxed{\textbf{9 cm}}$$

SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q8. An iron pillar has some part in the form of a right circular cylinder and remaining in the form of a right circular cone. The radius of base of each is 8 cm. The cylindrical part is 240 cm high and conical part is 36 cm high. Find the weight of the pillar which weighs 7.8 g per cm^3. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Total Volume of Pillar [2 Marks]

Common radius $r = 8$ cm, $H = 240$ cm, $h = 36$ cm.

Formula / Theorem Used: $V = \pi r^2 H + \frac{1}{3}\pi r^2 h = \pi r^2 \left(H + \frac{h}{3}\right)$

$$V = \frac{22}{7} \times 8^2 \times \left(240 + \frac{36}{3}\right) = \frac{22}{7} \times 64 \times 252 = 22 \times 64 \times 36 = 50,688 \text{ cm}^3$$

Step 2: Weight in kg (7.8 g/cm^3) [1 Mark]

$$\text{Weight} = 50688 \times 7.8 \text{ g} = 3,95,366.4 \text{ g} = \frac{395366.4}{1000} \text{ kg} = \boxed{\mathbf{395.3664} \text{ kg } (\approx \mathbf{395.37} \text{ kg})}$$

Q9. Find the volume of the largest right circular cone that can be cut out of a solid wooden cube whose edge is 9 cm. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Dimensions of Largest Cone [1 Mark]

Edge $a = 9$ cm $\implies$ max base diameter $= 9$ cm $\implies r = \frac{9}{2} = 4.5$ cm, height $h = 9$ cm.

Formula / Theorem Used: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$

Step 2: Arithmetic ($\pi = \frac{22}{7}$) [2 Marks]

$$V = \frac{1}{3} \times \frac{22}{7} \times \frac{9}{2} \times \frac{9}{2} \times 9 = \frac{11 \times 243}{14} = \frac{2673}{14} = \boxed{\mathbf{190\frac{13}{14}} \text{ cm}^3 \ (\approx \mathbf{190.93} \text{ cm}^3)}$$

SECTION D: Long Answer / HOTS Question — MATHS MANTHAN | (5 Marks)

Q10. Water in a canal, 6 m wide and 1.5 m deep, is flowing with a speed of 10 km/h. How much area will it irrigate in 30 minutes, if 8 cm of standing water is needed for irrigation? [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Water Volume Flowing in 30 Minutes [2.5 Marks]

Width $b = 6$ m, depth $h = 1.5$ m, speed $v = 10$ km/h $= 10,000$ m/h.
In 30 min $= \frac{1}{2}$ h, length of water column $L = 10,000 \times \frac{1}{2} = 5000$ m.

Formula / Theorem Used: $V_{\text{water}} = b \times h \times L =$ Irrigated Area × Standing Depth

$$V_{\text{water}} = 6 \times 1.5 \times 5000 = 9 \times 5000 = 45,000 \text{ m}^3$$

Step 2: Irrigated Area (8 cm = 0.08 m Standing Water) [2.5 Marks]

$$\text{Area} = \frac{45,000}{0.08} = \frac{45,00,000}{8} = \boxed{\mathbf{5,62,500} \text{ m}^2 \ (= \mathbf{56.25} \text{ hectares})}$$

Q11. Case Study: Sports Engineering – Surface Texture of a Dimpled Golf Ball

A tournament golf ball has a diameter of 4.2 cm (radius $R = 2.1$ cm). To reduce aerodynamic drag, its surface is indented with 315 identical hemispherical dimples, each having radius $r = 2$ mm $= 0.2$ cm. [4]

(i) Find the surface area of one hemispherical dimple. [1]

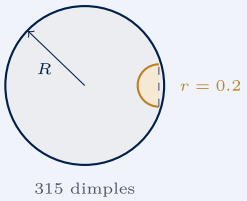
(ii) Find the volume of material carved out to create one dimple. [1]

(iii) Find the total volume of the golf ball before dimples were indented. [1]

(iv) Find the net surface area of the golf ball exposed to the surroundings. [1]

► CASE-BASED SOLUTION & MARKING SCHEME: [Total: 4 Marks]

Extracted Scenario Data: Ball radius $R = 2.1$ cm; 315 dimples of radius $r = 0.2$ cm $= \frac{1}{5}$ cm.



Formula / Theorem Used: $\text{CSA}_{\text{dimple}} = 2\pi r^2$, $V_{\text{dimple}} = \frac{2}{3}\pi r^3$, Net SA $= 4\pi R^2 + 315(\underbrace{2\pi r^2 - \pi r^2}_{\pi r^2})$

Sub-part (i): Surface Area of One Dimple [1 Mark]

$\text{CSA} = 2 \times \frac{22}{7} \times (0.2)^2 = \frac{44}{7} \times \frac{1}{25} = \frac{44}{175} \approx \boxed{\mathbf{0.2514\ cm^2}}$.

Sub-part (ii): Volume Carved Out per Dimple [1 Mark]

$V_1 = \frac{2}{3} \times \frac{22}{7} \times (0.2)^3 = \frac{44}{21} \times \frac{1}{125} = \frac{44}{2625} \approx \boxed{\mathbf{0.0168\ cm^3}}$.

Sub-part (iii): Total Ball Volume Before Dimpling [1 Mark]

$V_{\text{ball}} = \frac{4}{3} \times \frac{22}{7} \times (2.1)^3 = \frac{88}{21} \times 9.261 = 88 \times 0.441 = \boxed{\mathbf{38.808\ cm^3}}$.

Sub-part (iv): Net Exposed Surface Area [1 Mark]

Original SA $= 4 \times \frac{22}{7} \times (2.1)^2 = 88 \times 0.3 \times 2.1 = 55.44$ cm^2.

Net gain from 315 dimples $= 315 \times \frac{22}{7} \times (0.2)^2 = 45 \times 22 \times 0.04 = 39.60$ cm^2.

Net Exposed SA $= 55.44 + 39.60 = \boxed{\mathbf{95.04\ cm^2}}$.

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★